

Jongmin Choi

M.S. Student in Electrical Engineering · Multimodal AI (MMAI) Lab, KAIST

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RESEARCH INTERESTS

Biological intelligence provides the clearest evidence that learning can be both general and efficient, and I believe that building its principles into model architectures matters as much as scaling them up. From this perspective, I have studied multimodal learning that bridges the gap between audio and text while preserving the structure unique to each modality, as well as spiking neurons with temporal dynamics. Currently, guided by the idea that good reasoning begins with good input selection, I am exploring architectures that dynamically extract query-relevant information from diverse modalities and make it readily usable by language models. In parallel, I investigate the mathematical principles underlying deep network optimization.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea Mar 2025 – Present

M.S. in Electrical Engineering

Advisor: **Prof. Joon Son Chung** · Multimodal AI (MMAI) Lab · GPA **4.06 / 4.30**

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea Mar 2018 – Feb 2025

B.S. in Electrical Engineering and Mathematical Sciences (Double Major)

GPA **3.65 / 4.30**

PUBLICATIONS

** Equal contribution*

NeurIPS 2026 **FiTS: Interpretable Spiking Neurons via Frequency Selectivity and Temporal Shaping**

SPOTLIGHT

Jongmin Choi, Joon Son Chung

Neural Information Processing Systems · **0.95%** (292 / 30,709)

ICASSP 2026 **LAMB: LLM-based Audio Captioning with Modality Gap Bridging via Cauchy-Schwarz Divergence**

ORAL

Hyeongkeun Lee*, **Jongmin Choi***, KiHyun Nam, Joon Son Chung

IEEE International Conference on Acoustics, Speech, and Signal Processing

ICASSP 2026 **Diffusion-Link: Diffusion Probabilistic Model for Bridging the Audio-Text Modality Gap**

POSTER

KiHyun Nam*, **Jongmin Choi***, Hyeongkeun Lee, Jungwoo Heo, Joon Son Chung

IEEE International Conference on Acoustics, Speech, and Signal Processing

EMNLP 2026 **Fork-Merge Decoding: Enhancing Multimodal Understanding in Audio-Visual Large Language Models**

**FINDINGS
POSTER**

Chaeyoung Jung*, Youngjoon Jang*, **Jongmin Choi**, Joon Son Chung

Findings of Empirical Methods in Natural Language Processing

arXiv 2026 **OmniKVQuant: KV Cache Quantization for Omni-LLMs**

PREPRINT

Suho Yoo*, Hyunjong Ok*, **Jongmin Choi**, Jihoo Jung, Joon Son Chung

Under review at IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2027)

TEACHING EXPERIENCE

EE.70038 — Speech Recognition Systems — KAIST

Spring 2026

Teaching Assistant · Instructor: **Prof. Joon Son Chung**

EE.40034 — Deep Learning for Visual Understanding — KAIST

Fall 2025

Teaching Assistant · Instructor: **Prof. Joon Son Chung**

SKILLS

Research & ML: Python, PyTorch · **iOS Development:** Swift, SwiftUI, SwiftData · **Languages:** Korean (native), English (professional working proficiency)